Multi-Agent Case Report

Case: demo_case_ecg00001hr_mriPatient101_with_classifier

Contributing agents: ecg, mri

Note: This case pairs a PTB-XL ECG record with an ACDC MRI patient. These datasets share no real patients - this is a synthetic pairing for multi-agent architecture validation, not a real joint clinical case. The MRI patient (patient101, DCM) is a genuinely held-out test patient for the newly-integrated YOLO/U-Net/pathology-classifier pipeline - not used in any of that pipeline's training.

Integrated Summary (Sep Agent)

The ECG shows a heart rate of approximately 64 bpm with a mean RR interval of 939 ms and a QRS duration of around 60 ms. The MRI indicates low left ventricular ejection fraction at 29.9% and a high probability (96.5%) of dilated cardiomyopathy, along with signs suggestive of heart failure.

Flags for review: DCM, EF_low, low_lv_ejection_fraction_suspected, pathology_DCM_pattern_suspected

Traceability

Statement	Agent	Feature
The ECG reveals a mean heart rate of approximately 64 bpm	ecg	heart_rate_mean_bpm
The RR interval is around 939 ms	ecg	rr_mean_ms
The QRS duration is approximately 60 ms	ecg	qrs_duration_mean_ms
MRI shows a low left ventricular ejection fraction at 29.9%	mri	lv_ejection_fraction_pct
The MRI indicates a dilated cardiomyopathy with a high probability of 96.5%.	mri	pathology_DCM_probability

Note: traceability confirms each statement cites a real finding from a contributing agent - it does not verify that the summary's characterization of that finding (e.g. "low" vs "high") is clinically correct.

ECG Agent Findings

Record: 00001_hr

Flags: none

Feature	Value	Confidence	Evidence type	Rationale
duration_s	10.0	1.00	numeric_feature	Analyzed ECG trace duration.
qrs_count	11	1.00	numeric_feature	Number of detected QRS complexes.
heart_rate_mean_bpm	63.9	1.00	numeric_feature	Estimated mean heart rate.
heart_rate_std_bpm	1.4	1.00	numeric_feature	Standard deviation of heart rate.
rr_mean_ms	938.8	1.00	numeric_feature	Mean RR interval.
rr_std_ms	21.5	1.00	numeric_feature	Standard deviation of RR interval.
qrs_duration_mean_ms	59.3	1.00	numeric_feature	Mean QRS duration.
qrs_duration_std_ms	46.2	1.00	numeric_feature	Standard deviation of QRS duration.

MRI Agent Findings

Record: patient101

Flags: DCM, EF_low, low_lv_ejection_fraction_suspected, pathology_DCM_pattern_suspected

Feature	Value	Confidence	Evidence type	Rationale
lv_volume_ed_ml	222.6	1.00	numeric_feature	End-diastolic left ventricular volume.
lv_volume_es_ml	156.0	1.00	numeric_feature	End-systolic left ventricular volume.
rv_volume_ed_ml	129.8	1.00	numeric_feature	End-diastolic right ventricular volume.
rv_volume_es_ml	70.1	1.00	numeric_feature	End-systolic right ventricular volume.
myocardium_volume_ed_ml	127.6	1.00	numeric_feature	End-diastolic myocardium volume.
myocardium_volume_es_ml	128.8	1.00	numeric_feature	End-systolic myocardium volume.
lv_ejection_fraction_pct	29.9	1.00	numeric_feature	Left ventricular ejection fraction.
rv_ejection_fraction_pct	46.0	1.00	numeric_feature	Right ventricular ejection fraction.
pathology_DCM_probability	0.965	1.00	numeric_feature	Probability of dilated cardiomyopathy.
pathology_HCM_probability	0.019	1.00	numeric_feature	Probability of hypertrophic cardiomyopathy.
pathology_MINF_probability	0.0	1.00	numeric_feature	Probability of prior myocardial infarction with reduced function.
pathology_NOR_probability	0.015	1.00	numeric_feature	Probability of normal.
pathology_RV_probability	0.001	1.00	numeric_feature	Probability of abnormal right ventricle.
short_axis_mid_ed	short_axis_mid_ed	1.00	image_region	Short axis view of the heart in end-diastole.